



Flagline

Heat rules, handled on the sideline.

A free web app that helps every Texas school follow the new UIL heat rule for outdoor athletics and marching band: plan the practice, run the 30-minute rechecks, hear from athletes early, act fast in an emergency, and keep the record.

Flagline

Heat and smoke rules for **every outdoor practice.**

WBGT flags, 30-minute rechecks, athlete check-ins and the compliance log. Free, under the Texas UIL rule.

Texas UIL 2026-27 English and Spanish No hardware

ORANGE FLAG · NOW

87.6°

°F WBGT

- Max 2 hours of practice
- Cooling zone on site
- Next reading in 30 min

UIL Class 3 flag levels, °F WBGT

GREEN < 82.0°	YELLOW 82.0°	ORANGE 87.0°	RED 90.1°	BLACK 92.1°
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PROJECT DOSSIER

WarriorHacks 2.0 · Hackathon track

Theme: solve an issue in your community, county, state or nation

October 2026

TRY IT

flagline.codewiththeugene.top

.../coach/demo-pecan-creek-coach-2026

github.com/CodeWithEugene/WarriorHacks-2.0

The short version

The issue

From **August 1, 2026**, Texas UIL requires a wet bulb globe temperature (WBGT) reading before and every 30 minutes during every outdoor practice and marching band rehearsal, with limits on time, breaks and football gear at each flag level. Coaches and band directors, not trainers, usually run it. Paid systems cost hundreds to thousands of dollars per site. Exertional heat stroke is the leading preventable cause of death in high school football.

What we built

Flagline, a free, bilingual (English and Spanish) web app with no hardware and no accounts. It shows the flag now and for the next 7 days, answers plain-language plans, runs Practice Mode with timers and a meter photo reader, routes athlete check-ins with red-flag alerts, guides the heat stroke protocol and logs everything.

Why it is trustworthy

The safety core is deterministic code: a real physics model (Liljegren WBGT) checked against **288 reference cases**, UIL thresholds stored as versioned data with a conformance suite, **442 automated tests**. AI only interprets language and photos, gated on confidence. It never sets a flag level, a limit, or clears an athlete.

Status

Live in production at flagline.codewitheugene.top with a fictional demo team. Every feature in this dossier marked built works today on the live site. Open source under MIT.

Aug 1

2026 start of the UIL WBGT mandate

\$0

cost to schools; no hardware, no install

25/25

red-flag check-ins routed to emergency in our eval, 0 of 8 false alarms

2

languages across every screen (462 messages each)

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A new rule, a deadly risk, and no one to run it

Every afternoon from August to October, someone decides whether and how a team practices in dangerous heat. The right measure is WBGT, which combines air temperature, humidity, wind and sun. It is not the number on a weather app.

UIL, 2026-27: “The use of Wet Bulb Globe Temperature (WBGT) to monitor environmental conditions and guide activity modifications is no longer a recommendation, but a required standard for all UIL outdoor athletic and marching band activities.” Readings are required within 15 minutes before practice and every 30 minutes during it.

The stakes

#1

Exertional heat stroke is the leading preventable cause of death in youth, high school and college football, nearly 3 deaths per season

NFHS SMAC task force, 2026

9,237

time-loss heat illnesses among US high school athletes each year; football's rate is 10 times other sports

CDC MMWR 59(32)

100%

survival when cooling starts within 10 minutes of collapse. The order matters: cool first, transport second

UIL resources

67

exertional heat stroke deaths in US secondary school sports, 1982 to 2022; 94% in football

Stearns et al., Sports Health 2025

~80%

of Texas coaches surveyed call the WBGT requirement too restrictive; there is no UIL requirement to employ a trainer

Houston Public Media, Aug 2026

2/7

Texas score on heat acclimatization policy; 0/3 on cold-water immersion before transport

Korey Stringer Institute, 2025-26

It is local

A 15-year-old Lancaster (Dallas area) player died of heat stroke in August 2025 after conditioning in a non-air-conditioned facility. An Austin Bowie High School player spent time in the ICU after heat stroke in a 2023 game. Band is at risk too: about 400 marching band members with exertional heat illness were reported in the news from 1990 to 2020, with no national band standard, and a university band member died after collapsing at a 91 F game on September 5, 2026.

It is national

A record 8,266,244 US high school students played sports in 2024-25 (NFHS). Texas joins at least 17 other states that require WBGT-guided practices, and states have adopted nearly 200 heat-illness prevention policies since 2017. Parents estimate their kids lost about a week of practices and games in 2024 to heat, smoke and other weather.

Who it hurts, and why it keeps happening

Person	Their day	What hurts today
Head coach	Teaches all day, runs practice 3:45 to 6:00 pm for 60 to 120 athletes	Checks a weather app and guesses; forgets 30-minute rechecks when practice gets busy; no records
Band director	Rehearses 150+ students on a hot parking lot, 4:00 to 7:00 pm, Sep to Nov	UIL rules now cover band; band culture is less used to heat protocols; often no trainer on site
Athletic trainer	One trainer for several sports on several fields	Cannot be at every field; needs every field's status and every athlete red flag in one place
Athletic director	Oversees compliance for a campus	Needs evidence that every practice followed the protocol
Student athlete	14 to 18 years old, practicing in the heat	Pressure not to complain; no private, low-friction way to say "I feel wrong"
Parent	Gets cancellation texts	No view of the conditions or the reasons behind decisions

The routine today, and where it breaks

Step	What should happen	What actually goes wrong
Planning	Know the expected flag and move practice or change gear	Weather apps show air temperature or heat index, not WBGT
Pre-practice	A reading within 15 minutes before the start	Nobody owns it; skipped or taken in the wrong spot
Applying the rules	Right class (Class 2 or 3), right row, gear and cooling limits	Wrong class, wrong row, gear limits forgotten
During practice	A reading every 30 minutes; switch to stricter rules at once	Forgotten in the flow of coaching 100 teenagers
Warning signs	Recognize early; cool first, then call 911	Late recognition; transport before cooling
Records	Keep WBGT records on file	A clipboard lost in a locker room, or nothing
Smoke days	Check AQI in the same decision	A separate app, often not checked

Root causes

Wrong tool. Weather apps do not show WBGT or know sport rules.

Tables, not workflows. No one turns the chart into today's timeline.

Cost. Meters cost \$549+; site systems about \$3,500 per site per year.

No loop for symptoms. Students under-report; adults miss early signs.

Load at the worst time. Rechecks compete with coaching in sun and noise.

No records by default. Evidence of compliance is an afterthought.

Tools give a number. None runs the practice.

	Flagline	UNC WBGT forecast (UIL recommends)	Zelus WBGT	Perry Weather	Kestrel 5400 meter
Price	Free	Free	\$49.99 to \$299 per year (free tier ended Jun 2026)	About \$3,500 per site per year	\$549 to \$629
Hardware	None (meter optional)	None	None	Required	Is the hardware
WBGT source	Two independent forecasts plus on-site readings	Forecast twice daily	Proprietary; read 2 to 3 C cool when hot (KSI 2025)	On-site sensor	Measured
Exact UIL Class 2 and 3 rules	Yes, tested	Plot bands; banner labels mismatch	Selectable	Yes	No
Sport outputs (football gear, band)	Yes	No	Partial	Partial	No
Practice workflow and timers	Yes	No	Partial	Alerts	No
Log of readings and decisions	Yes, free (CSV)	No	Paid tier	Paid tier	With kit
Air quality in the same view	Yes	No	Top tier	Add-on	No
Athlete check-ins and red flags	Yes	No	No	No	No
Heat stroke protocol	Yes, with cooling timer	No	No	No	No
Plain-language planning	Yes (TypeSafe Jev)	No	No	No	No
English and Spanish	Yes	No	Unknown	Unknown	No

The gap. Nothing free says: “At 4:30 pm on this field, under your state's rules, for this team in full pads, you may practice 2 hours with 4 breaks of 4 minutes; cooling tubs must be filled; your next reading is due at 5:00”, and then keeps the record by itself.

Positioning

For Texas high school coaches and band directors who must follow the new UIL heat rule without a trainer at every practice, Flagline is a free web app that turns the WBGT forecast and on-site readings into a practice plan, a recheck routine and a record. Unlike the free UNC forecast, Flagline runs the practice. Unlike paid systems, it needs no hardware and no budget.

Not built for professional or college programs with full sports medicine staffs, individual outdoor workers (OSHA and NIOSH tools exist), or anyone expecting a medical diagnosis.

Five verbs: plan, prepare, run, protect, prove

Plan

The WBGT flag for every practice hour for 7 days, and plain-language questions like “Can we go full pads 4 to 6 tomorrow?” answered with a rules check and better time windows. **Quick Check** works for any place: UIL rules in Texas, Korey Stringer Institute categories elsewhere.

Prepare

Exactly what the rules require at the expected level: maximum length, breaks per hour, football gear, conditioning, cooling zone, water. Borderline readings (within 1 F of the next level) are flagged so the stricter preparations are ready.

Run

Practice Mode: the pre-practice reading (keypad or a photo of the meter), a 30-minute recheck countdown, a break timer, time used against the level's limit, a cooling zone checklist, a thunder hold, and instant rule changes when the level rises. The coach and staff see the same practice, synced.

Protect

Athletes scan a QR code and say how they feel, in English or Spanish, with no account. Red flags (confusion, stopped sweating, fainting) alert the coach at once and open the **heat stroke protocol**: cool first, transport second, with a cooling timer and temperature log.

Prove

Every reading, level change, break and check-in is timestamped in a live log and exportable as CSV for the athletic director. Parents get a public team page.

Principles we did not bend

The human decides. The app never says “safe”. It shows the flag and what the rules require.

Honest numbers. Every value shows its source and whether it is a forecast or a measurement.

Five seconds on the sideline. Every live action is one or two taps, with 44 px targets.

Bilingual from day one. English and Spanish everywhere, including check-ins.

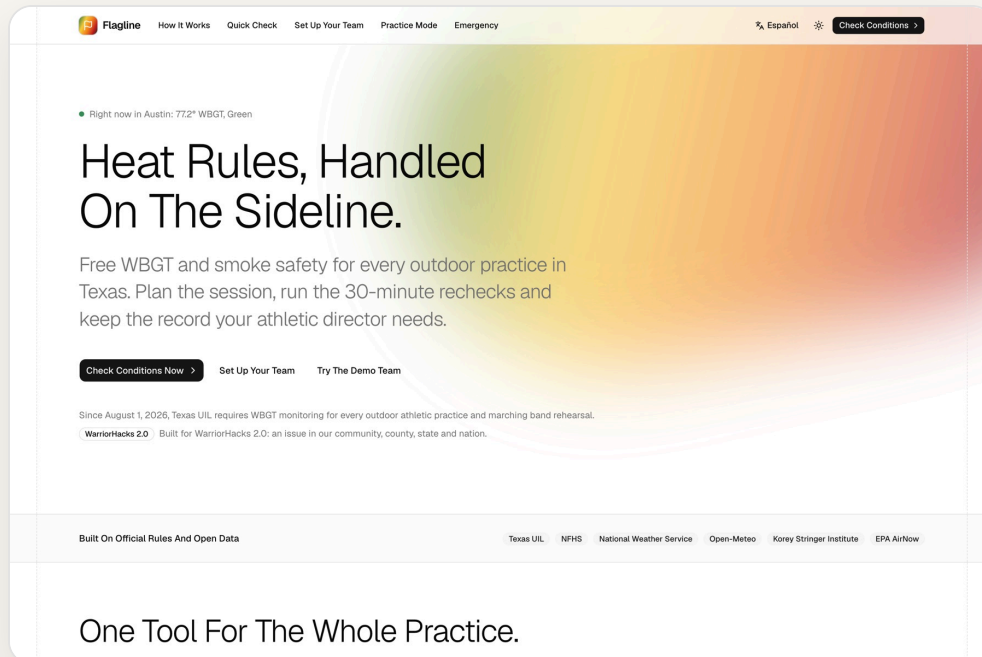
Deterministic where it matters. Heat math and rules are tested code; AI never decides a level.

The measured reading wins. Forecasts plan; on-site readings drive the live rules.

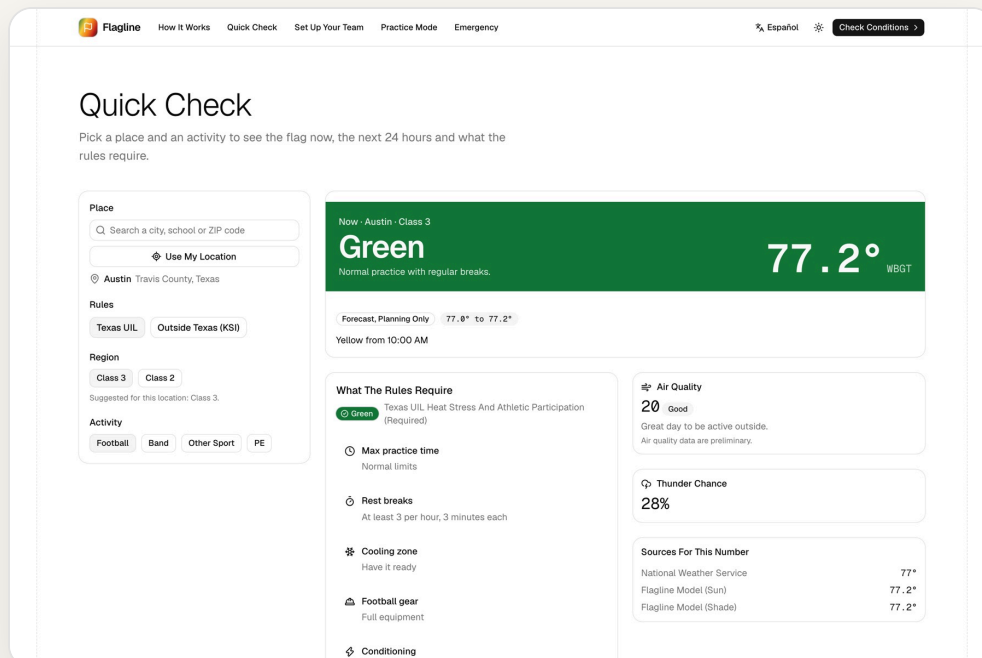
Color is never the only signal. Every flag has an icon, a name and a value.

Privacy by default. No athlete names needed; check-ins can be anonymous.

Know the flag before you reach the field

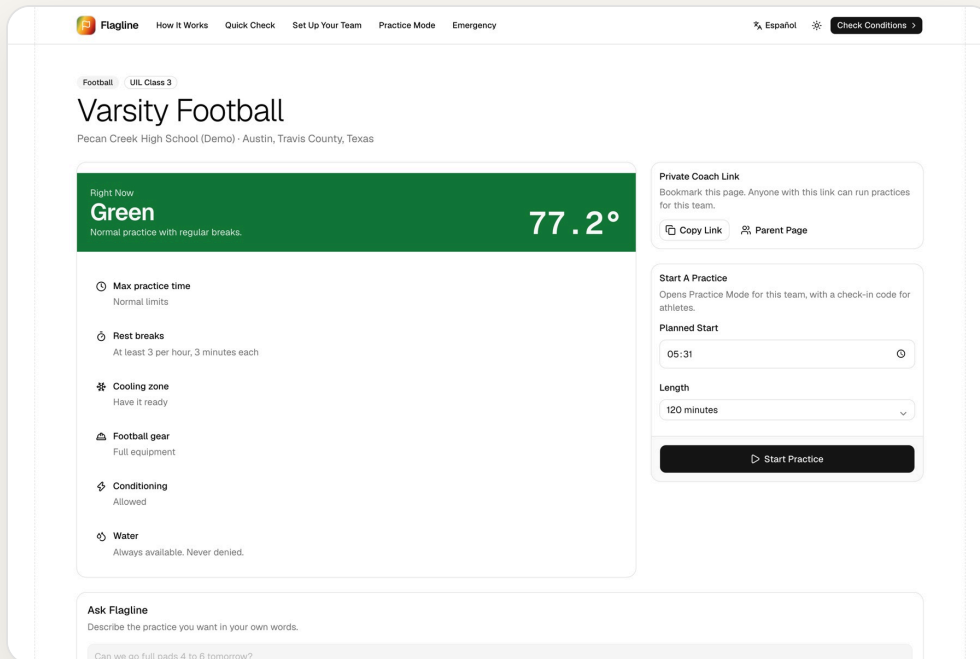


Home. The live Austin flag, the promise in one line, and three ways in: Check Conditions, Set Up Your Team, Try The Demo Team. Built on official rules and open data (UIL, NFHS, NWS, Open-Meteo, KSI, EPA AirNow).

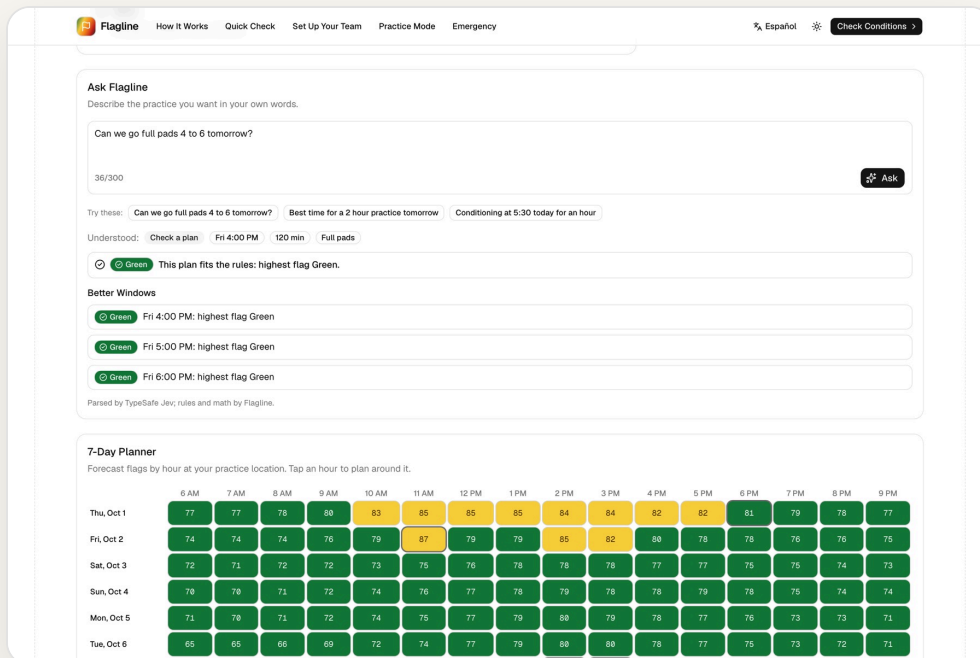


Quick Check. Pick a place and an activity. Flagline shows the flag now, the next 24 hours and exactly what the rules require, with air quality, thunder chance and the sources behind the number. It suggests the right UIL class for the location, and uses KSI categories outside Texas.

Ask in plain words. The rules do the math.

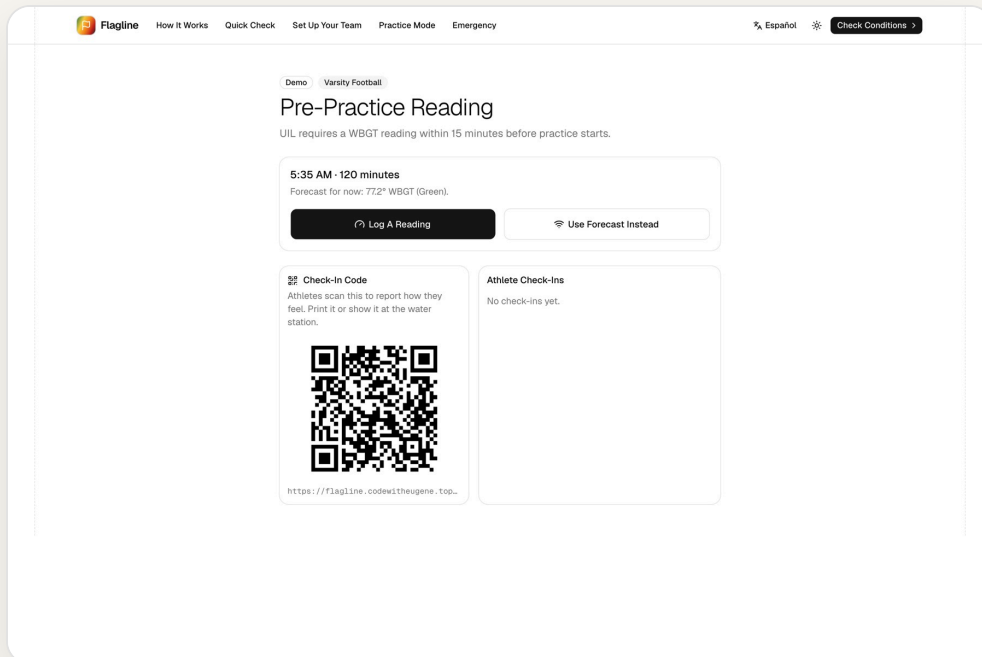


Coach dashboard. A team gets a private coach link (no accounts). Today's flag and requirements, a one-tap Start Practice, and a parent page.

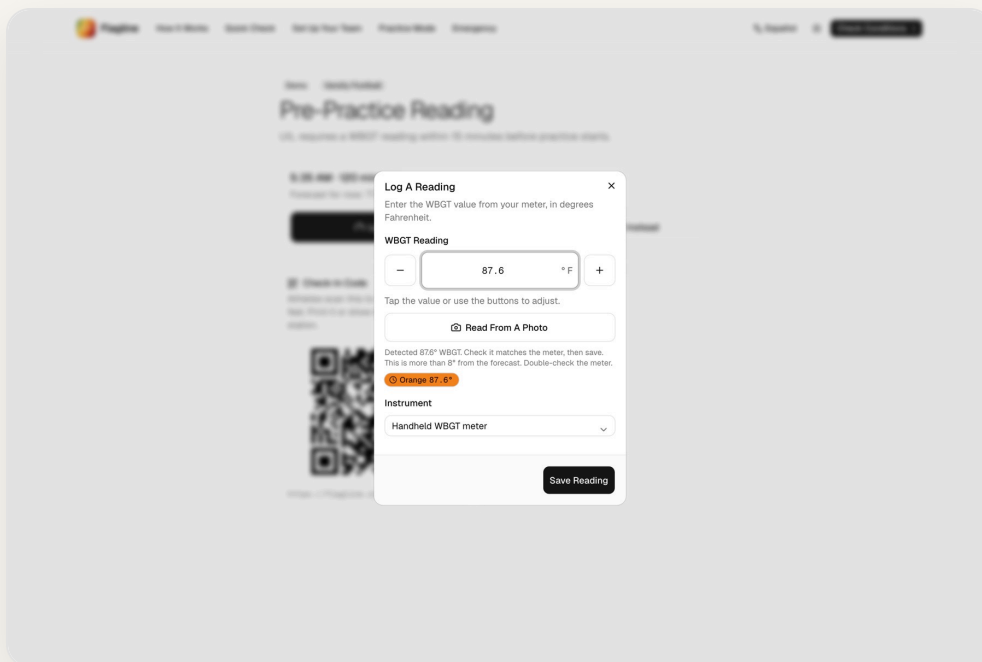


Ask Flagline and the 7-day planner. “Can we go full pads 4 to 6 tomorrow?” TypeSafe Jev reads the intent, gear and times; the deterministic rules engine checks the plan hour by hour and suggests better windows. The planner shows every practice hour for a week.

A reading in five seconds, even from a photo

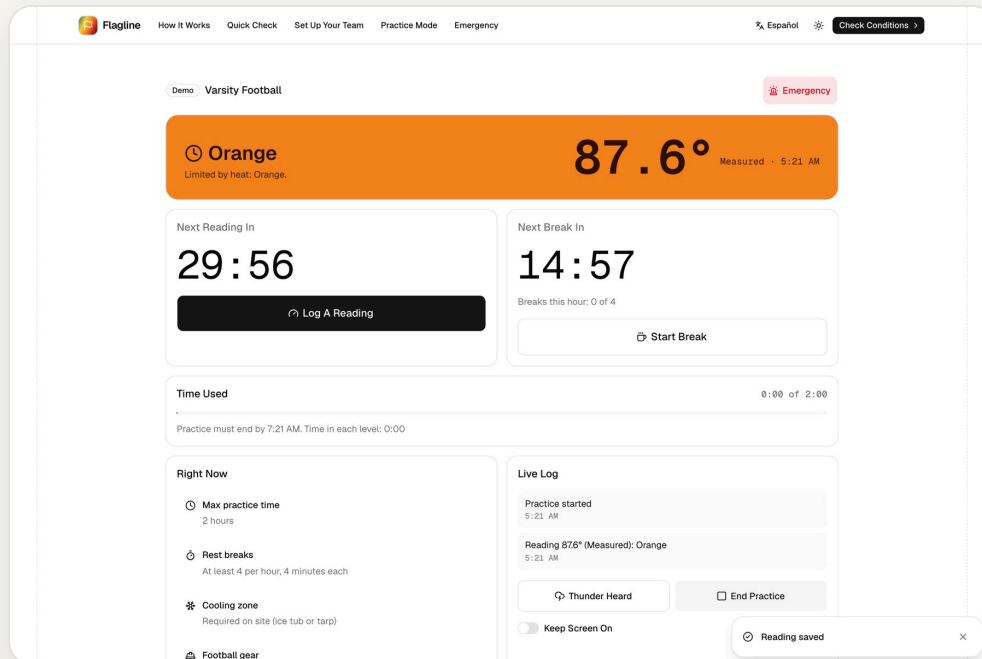


Pre-practice reading. UIL requires a reading within 15 minutes before practice. One tap to log a meter value or use the forecast, with a QR code athletes can scan at the water station.

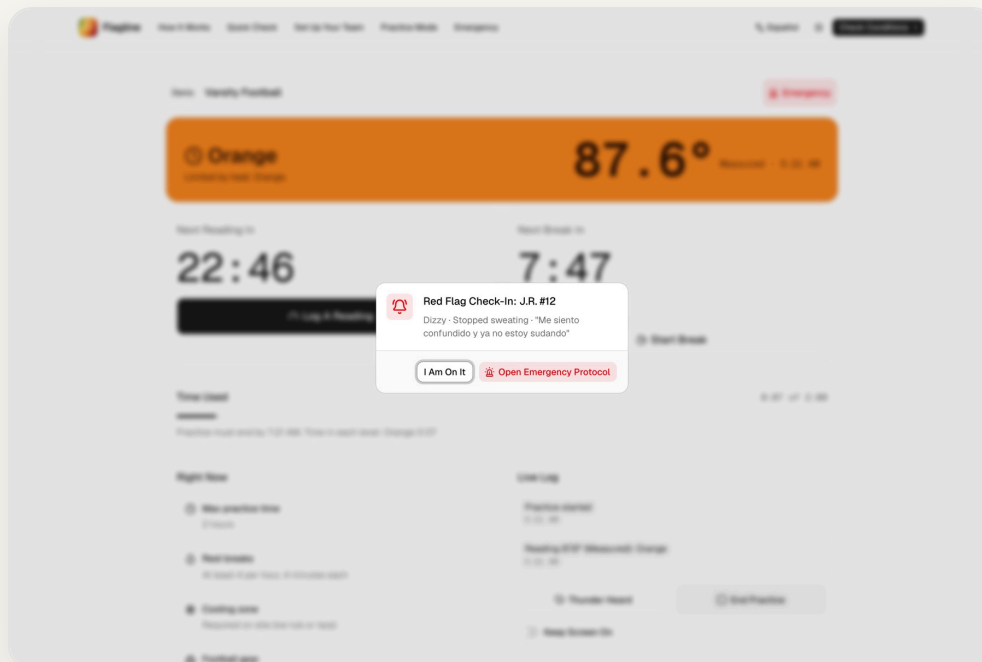


Meter photo. A vision model transcribes the display, Jev picks which number is WBGT (even when the screen shows several), code checks the range and warns when it is far from the forecast. The coach always confirms before saving.

The routine runs itself. Red flags cannot wait.



Practice Mode. Orange flag at 87.6 F: two hours maximum, four breaks of four minutes every hour, cooling zone required. A countdown to the next reading, a break timer, time used against the limit, a thunder hold and a live log.



Red flag alert. When an athlete reports a red flag, the coach's screen interrupts with the message and one tap to the emergency protocol.

Athletes get a voice. Coaches get the order right.

Athlete Check-In

Athletes speak up in their own words.

Scan the code at the water station, tap symptoms or type, in English or Spanish. No account needed.

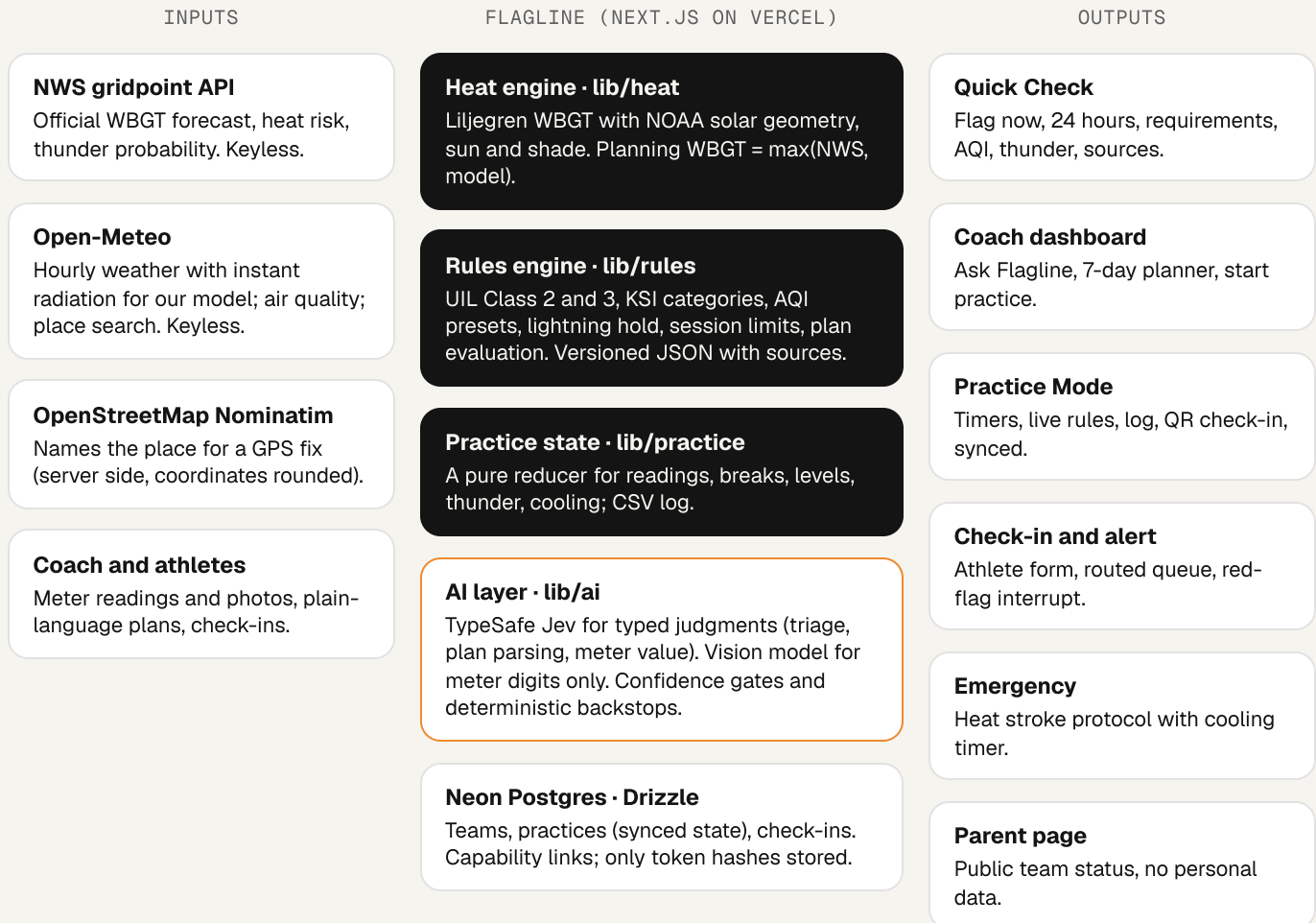
1. Code rules catch tapped red flags and warning words.
2. TypeSafe Jev catches what the words alone miss.
3. Red flags alert the coach and open the protocol.

Athlete check-in. No account. Tap symptoms or type, in English or Spanish. “Me siento confundido y ya no estoy sudando” (I feel confused and I have stopped sweating) routes to Emergency in seconds, caught both by our keyword rules and by Jev.

Queue. Open emergencies first, with plain-language reasons for each flag.

Protocol. Cool first, transport second: a cooling timer, Call 911, temperature log, EMS times.

Deterministic core, AI only at the edges



Stack

Next.js 16 (App Router, Turbopack) · React 19 · TypeScript · Tailwind CSS v4 · pure shadcn/ui (base-nova on Base UI) · Zod 4 · next-intl (English and Spanish) · Recharts · chrono-node · Drizzle ORM with Neon Postgres · TypeSafe Jev via @typesafe-ai/sdk · OpenRouter vision (Kimi K3) · Vitest · Vercel hosting behind Cloudflare · pnpm.

Design choices that matter

- **No accounts yet, on purpose.** A coach link (only its SHA-256 hash is stored), a check-in token and a public slug. Nothing personal is required to use it.
- **Two independent forecasts.** The flag always uses the higher of NWS and our model; outside the US, the model runs alone and says so.
- **Server-only AI keys,** rate limits on every public endpoint, Zod validation at every boundary, security headers.

Real physics, the exact UIL table

WBGT, computed

Outdoor WBGT = $0.7 \times \text{natural wet bulb} + 0.2 \times \text{globe} + 0.1 \times \text{air}$. We ported the Liljegren et al. (2008) model from ECMWF's open-source thermofeel library to TypeScript, including the globe and natural wet bulb solvers, 2 m wind adjustment and NOAA solar geometry. It matches **288 reference cases** generated from the original implementation. On a late-September Austin afternoon: air 97 to 99 F, heat index 100 to 107 F, WBGT 85 to 88 F.

UIL WBGT activity guidelines, 2026-27

Level	Class 3 (most of Texas)	Class 2 (Panhandle, South Plains)	What the rules require
Green	< 82.0	< 79.7	Normal activities; at least 3 rest breaks each hour, 3 minutes each
Yellow	82.0 to 86.9	79.7 to 84.6	Discretion for intense or prolonged exercise; 3 breaks of 4 minutes each hour; cooling zone required
Orange	87.0 to 90.0	84.7 to 87.6	2 hours maximum ; football in helmet, shoulder pads and shorts (pants allowed if reached mid-practice); 4 breaks of 4 minutes each hour
Red	90.1 to 92.0	87.7 to 89.7	1 hour maximum ; football: no protective equipment and no conditioning; 20 minutes of breaks across the hour
Black	≥ 92.1	≥ 89.8	No outdoor workouts ; delay until cooler

Timing rules we enforce

- A reading within 15 minutes before practice and every 30 minutes during it (with a 2-minute grace).
- When conditions worsen, everyone moves to the stricter level at once; when they improve, practice is not extended.
- Water is never denied; breaks mean unlimited hydration and no activity.
- Thunder or lightning starts a 30-minute hold that resets with each strike (UIL adopts NFHS).
- Air quality presets: modify at AQI 101 to 150, cancel youth activity at 151+, with a stricter youth option.

Outside Texas

Korey Stringer Institute regional categories (Grundstein et al. 2015) apply anywhere else, with Category 1 as the most protective default.

Typed judgments, not chatbot text

Instead of prompting a chatbot and parsing its words, Flagline asks **TypeSafe Jev** typed questions (Choice, Noul probabilities and Scores) with confidence values our code can gate on. AI handles only what code cannot: understanding people and photos.

Check-in triage

Jev screens free text for 8 red flags (confusion, collapse, vomiting, stopped sweating, severe headache, seizure, breathing, chest pain) plus severity and category, in one request.

Routing is recall-first and deterministic: any tapped red flag, an English or Spanish keyword, a red-flag probability ≥ 0.25 , severity ≥ 3.0 or “possible heat stroke” goes to Emergency. AI can raise a flag, never clear one.

Ask Flagline

A date parser proposes start times and lengths (“4 to 6 tomorrow” means 4 PM; “about 2 hours” is a length). Jev picks intent, gear, conditioning and the right candidate in one request.

Below 0.6 confidence the app asks instead of guessing. The rules engine then evaluates the plan hour by hour and finds better windows.

Meter photo

A vision model (Kimi K3 via OpenRouter) only transcribes the numbers and labels on the display. Jev picks which number is WBGT.

Code converts Celsius, rejects values outside 40 to 120 F, warns when more than 8 F from the forecast, and the coach confirms. Photos are read once and never stored.

Measured results

Eval	Result	What it means
Red-flag triage, 25 messages (English and Spanish)	25/25 routed to Emergency	100% recall, the bar we set; 8 of them were caught only by Jev, not by keywords
Routine messages, 8	0/8 false alarms	Recall-first without crying wolf on cramps or tiredness
Meter photos, 4 screens	4/4 read correctly	Including several readings on one screen, a Celsius display and a blurred, tilted dark screen
Ask Flagline parsing	Unit tested edge cases	PM inference, duration phrases, same-day times, Spanish dates, intent settled by a stated time

Safety contract. AI never sets a flag level, never sets a practice limit, and never tells anyone an athlete is fine. Every AI-derived value is shown and editable before it is used.

Built for the worst afternoon of the season

Hazards and controls

Hazard	Controls
App shows a lower level than reality	Higher of two forecasts; borderline flag within 1 F; forecast labeled planning only; on-site reading prompted before and every 30 minutes; the measured reading drives the rules
A rule encoded wrongly	Rules are versioned data with source links; conformance tests on every boundary; a threshold change needs a source and a passing test
Missed recheck	Big countdown, overdue state, synced practice for staff, logged
An athlete's red flag not escalated	Tapped red flags bypass AI; keyword backstop in two languages; recall-first thresholds; 100% recall eval; the app never clears a flag
Transport before cooling	The protocol leads with cooling; Call 911 after cooling starts; large type; bilingual
Over-trust	Never says "safe"; the coach confirms; copy frames the app as support for school staff
Misread meter photo	Range and forecast checks; the coach confirms; manual entry always available

Privacy

- No athlete names required; jersey numbers or initials are optional; check-ins can be anonymous.
- No accounts; coach links stored only as hashes.
- Meter photos processed in memory and never stored.
- Public team pages hold no personal data.
- All demo data is fictional (Pecan Creek High School).
- AI keys stay on the server.

Accessibility

- Every flag is color plus icon plus name plus value.
- 44 px targets in Practice Mode; one-handed layouts.
- Light, dark and a sunlight theme for glare.
- Labels match visible text (WCAG 2.5.3); selected states never rely on shade alone.
- English and Spanish on every screen, enforced by a test that checks every message key.

Flagline is not a medical device and does not diagnose illness. It does not replace an athletic trainer, an on-site WBGT instrument or emergency services. Decisions remain with school staff under the UIL heat plan and the school's emergency action plan.

Tested like it matters

442

automated tests passing
across 11 test files

288

WBGT reference cases
matched (thermofeel)

100%

UIL conformance suite
on every level boundary

462

messages in each
language, parity checked

What the tests cover

Area	Checks
Heat physics	Globe and natural wet bulb solvers, 2 m wind, solar zenith, sun and shade WBGT against the reference fixture
UIL and KSI rules	Every threshold edge for Class 2 and 3, KSI categories, requirements by level and sport, mid-practice gear, borderline detection
Conditions	NWS duration parsing and unit conversion, Open-Meteo parsing, two-source combination, reverse geocoding
Practice state	Readings, starts, breaks per hour, level changes, thunder hold, cooling, end, CSV log
Planning	Plan evaluation hour by hour, best windows, date and duration parsing, intent gate
AI routing	Triage thresholds, keyword lists in English and Spanish, third-party distress, meter transcription and selection
Check-in queue	Emergencies first, newest first within a level, readable reasons
Messages	English and Spanish keys match; every key the app asks for exists; no em or en dashes in copy

Live evals (run against the real models)

`pnpm eval:triage` · `pnpm eval:ask` · `pnpm eval:meter`. Results are on page 14. Rule thresholds can only change with a source and a passing conformance test.

Engineering hygiene

- TypeScript strict, ESLint and a production build checked before every deploy.
- Zod validation at every API and Server Action boundary; rate limits on public endpoints.
- Security headers (camera and geolocation limited to the app itself); secrets only in environment variables.
- Pure shadcn/ui components from the default registry; no other UI libraries.

Free to run, easy to adopt, ready to spread

Who benefits first

Every UIL member high school and junior high with outdoor athletics or a marching band, starting with the schools that have no trainer at every practice and no hardware budget. Nationally, 8.27 million high school athletes and the 17+ states that already require WBGT-guided practices.

How we will measure impact

Outcome	Metric	How
Faster, correct planning	Time from opening to the practice plan (target under 10 s)	Privacy-preserving product analytics
Compliance	Rechecks on time; modifications confirmed	Practice logs
Earlier symptom reporting	Check-ins per practice; time to adult response	Anonymous check-in records
Fewer lost practices	Practices moved instead of cancelled	Planner and log
Safety	Cooling started within 10 minutes in any incident	Incident records, if any

Long term: fewer heat illnesses and zero heat deaths in participating programs. That cannot be shown in two weeks, and we say so.

Operating cost at pilot scale (about 50 schools)

\$0

weather and air data
(NWS public domain,
Open-Meteo, keyless)

\$0

hosting and database on
free tiers (Vercel, Neon)

< \$1

per month for Jev
judgments

< \$30

per month all in,
including meter photos

Forecasts are cached by grid cell and coordinates, so every team at a school shares the same calls.

Adoption path

1. **Pilot** with 3 to 5 Austin-area schools this fall, measured against the metrics above.
2. **Channels**: Texas High School Coaches Association, Texas Music Educators Association for band, district athletic directors and trainer networks. UIL already points coaches to internet-based WBGT sources.
3. **Model**: free forever for schools; any later district tier never gates a safety feature.
4. **Beyond Texas**: each new state is a rules data file plus conformance tests, not a rewrite.
5. **Open source** (MIT) so districts can self-host and audit the rules.

How Flagline answers the four criteria

Impact

A life-safety problem with official, current evidence; a brand-new mandate in the judges' own state; a named audience (every Texas school with outdoor athletics or band); equity for schools without trainers or budgets; a nation-wide path.

Feasibility

Already live on free infrastructure with free public data, no hardware and no accounts. Under \$30 per month for dozens of schools. A clear pilot and adoption plan; rules for new states are data.

User experience

One glanceable flag with text, not color alone; a big recheck timer; a five-second reading by keypad or photo; plain-language planning; English and Spanish; sunlight mode; 44 px targets; the athlete form needs no account.

Technical craft

A ported physics model with 288 reference cases; a typed, tested rules engine; two-source uncertainty; confidence-gated typed AI with deterministic backstops; synced practices; 442 tests and live evals.

The theme: community, county, state, nation

Scale	How Flagline serves it
Community	The teams, bands and families of Austin, starting with the fictional Pecan Creek High School demo team
County	Travis County schools, where late-September afternoons still reach orange and red WBGT
State	Texas made WBGT monitoring mandatory for every outdoor practice and marching band rehearsal from August 1, 2026; Flagline implements the exact UIL rule, Class 2 and Class 3
Nation	At least 17 other states require WBGT-guided practices; Quick Check already works anywhere with KSI categories, and new states are a data task

Texas made heat safety mandatory for every outdoor practice this season. Flagline helps every school in our community, county and state follow it, free, so no student athlete dies of a preventable heat stroke.

What ships today, and what comes next

Built and live

- ✓ Quick Check worldwide, with GPS fix accuracy and place names
- ✓ Two-source WBGT (NWS plus Liljegren model) with ranges
- ✓ UIL Class 2 and 3 plus KSI categories; AQI; thunder hold
- ✓ Teams with private coach links and parent pages
- ✓ Ask Flagline with Jev and the 7-day planner
- ✓ Practice Mode: timers, breaks, live rules, cooling checklist, log, CSV
- ✓ Synced practices for the coaching staff
- ✓ Meter photo reading (vision plus Jev plus checks)
- ✓ Athlete check-ins in two languages with Jev triage
- ✓ Red-flag alert, sorted queue, readable reasons
- ✓ Heat stroke protocol with cooling timer
- ✓ Light, dark and sunlight themes; English and Spanish

Next

- A signed, bilingual PDF practice log for athletic directors
- Push notifications for rechecks and red-flag check-ins
- Offline Practice Mode for fields with weak signal
- School accounts with roles (owner, coach, trainer, viewer)
- Football acclimatization tracking per athlete
- AI-assisted import of other states' heat policies
- Forecast calibration from logged on-site readings
- Automated lightning data with the 10-mile rule

Build log

Date	Milestone
Sep 25	Research and planning documents only (challenge, problem, solution, build spec, sources). No application code before the theme reveal.
Sep 27	Theme revealed; Flagline confirmed by our theme-fit protocol (Jev rated it 3.77 of 4, the highest of five candidates).
Sep 28	Scaffold (Next.js 16, shadcn/ui base-nova); WBGT physics, UIL rules engine, two-source conditions; Home, Quick Check, Practice Mode, Emergency.
Sep 28 to 29	Teams, synced Practice Mode, athlete check-ins with Jev triage, Ask Flagline; deployed to production with a custom domain.
Sep 30	Meter photo reading; worldwide location with reverse geocoding and fix accuracy.
Oct 1	Polish from end-to-end testing: queue ordering, readable flag reasons, accessibility fixes, message-key test; submission media and demo video.

Open about how it was made

AI disclosure

Claude Code assisted with writing code and documentation. In the product, TypeSafe Jev makes typed judgments (triage, plan parsing, meter value selection) and Kimi K3 via OpenRouter transcribes meter photos. The demo video was assembled with OpenMontage and Remotion, narrated by a synthetic voice (Microsoft en-KE Chilemba).

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WBGT method: Liljegren et al. (2008), ported from ECMWF thermofeel (Apache-2.0). Weather: National Weather Service (public domain) and Open-Meteo (CC BY 4.0). Place names: OpenStreetMap contributors. UI: shadcn/ui, lucide icons, Geist fonts. Music: Pixabay Content License. Flagline is MIT licensed.

Planning documents were written before the theme reveal; all application code was written during the event. Demo data is fictional; no real minors' data appears anywhere.

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